

Hemoglobin Disorder Classification Using Machine Learning and Neural Networks

TECHNICAL PROJECT REPORT & MODEL EVALUATION

Introduction

This project focuses on classifying hemoglobin disorders using Machine Learning and Neural Network models. The dataset contains medical and laboratory information related to blood disorders. The main goal of the project is to compare different classification models and determine which model performs best on this dataset.

Dataset Description

The dataset contains 3000 rows and 34 columns. The target variable is `variant_type`, which represents different types of hemoglobin disorders.

The dataset includes:

- Clinical indicators
- Blood measurements
- Hemoglobin-related features
- Binary medical conditions

Note: No missing values were found in the dataset. However, the dataset is imbalanced because one class contains significantly more samples than the other classes.

Data Preparation

Several preprocessing steps were applied before training the models:

- Separating features and target variable
- Encoding categorical variables
- Splitting the dataset into training and testing sets
- Applying feature scaling using `StandardScaler`

The dataset was divided into:

- 80% training data
- 20% testing data



Machine Learning Models

Three Machine Learning models were trained and evaluated:

1. Logistic Regression

Logistic Regression achieved strong performance with good overall classification results. However, it performed better on the majority class than the minority classes.

2. Random Forest

Random Forest achieved the best overall performance among all traditional Machine Learning models. It produced the highest Accuracy and F1-score.

3. K-Nearest Neighbors (KNN)

KNN achieved good results, but its performance was lower than Random Forest. The model was more affected by the class imbalance problem.



Neural Network

A Neural Network model was built using TensorFlow and Keras.

The model included:

- Dense layers
- ReLU activation function
- Softmax output layer
- Dropout layers to reduce overfitting

The model was trained for 15 epochs using training and validation data. Training and validation curves were plotted to monitor the model's learning process.

The Neural Network achieved strong performance, but Random Forest still produced slightly better results on this dataset.



Results and Comparison

The models were evaluated using the following metrics:

Accuracy

Precision

Recall

F1-score

Final comparison results showed that:

- Random Forest achieved the highest overall performance

- Neural Network achieved strong results
- KNN achieved the lowest performance among the tested models

The class imbalance affected the performance of some models, especially when predicting minority classes.

What I Learned

Through this project, I learned how to:

- Prepare and preprocess datasets
- Train and evaluate Machine Learning models
- Build Neural Networks using TensorFlow and Keras
- Compare models using classification metrics

I also learned that more complex models are not always better, since Random Forest achieved better performance than the Neural Network on this dataset.

Conclusion

This project successfully applied Machine Learning and Neural Network techniques to classify hemoglobin disorders using medical data.

Among all tested models, Random Forest achieved the best overall performance and proved to be the most effective model for this dataset.